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检索结果：

数据库	检索结果
SCIE（第一作者论文收录篇数）	2 篇
SCIE（非第一作者论文收录篇数）	6 篇
EI （论文收录篇数）	1 篇

附件一：SCIE、EI 收录论文详单

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2020 年 9 月 25 日

标题: Optimal Representative Distribution Margin Machine for Multi-Instance Learning

作者: Luan, TX (Luan, Tianxiang); Luo, TJ (Luo, Tingjin); Zhuge, WZ (Zhuge, Wenzhang); Hou, CP (Hou, Chenping)

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摘要: Multi-instance learning (MIL) plays an important role in many real applications, such as image recognition and text classification. The instance-based approach selects instances in each bag to train and has drawn significant attention recently. However, less work took the distribution information in the account and the margin distribution has been proven to be important to the generalization performance. In this paper, we propose an optimal representative distribution margin approach for multi-instance learning (MIORDM). The representative instances are the samples from the instance space and the distribution of them is important for us to find the best separation hyperplane. As the representative instances are selected iteratively, in each iteration, the instances will be more precise by the best hyperplane and the model will be better in the next iteration. In this way, a well-performed method can be derived with better generalization performance. Experiments compared with other types of state-of-the-art approaches on different datasets show that our method outperforms the others and achieves better generalization performance.

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文献类型: Article

作者关键词: Support vector machines; Training; Image recognition; Supervised learning; Task analysis; Predictive models; Linear programming; Multi-instance learning; optimal margin distribution machine; representative instance

KeyWords Plus: INTELLIGENCE; TESTS

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第 3 条，共 8 条

标题: Dimension Reduction for Non-Gaussian Data by Adaptive Discriminative Analysis

作者: Luo, TJ (Luo, Tingjin); Hou, CP (Hou, Chenping); Nie, FP (Nie, Feiping); Yi, DY (Yi, Dongyun)

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摘要: High-dimensional non-Gaussian data are ubiquitous in many real applications. Face recognition is a typical example of such scenarios. The sampled face images of each person in the original data space are more closely located to each other than to those of the same individuals due to the changes of various conditions like illumination, pose variation, and facial expression. They are often non-Gaussian and differentiating the importance of each data point has been recognized as an effective approach to process the high-dimensional non-Gaussian data. In this paper, to embed non-Gaussian data well, we propose a novel unified framework named adaptive discriminative analysis (ADA), which combines the sample's importance measurement and subspace learning in a unified framework. Therefore, our ADA can preserve the within-class local structure and learn the discriminative transformation functions simultaneously by minimizing the distances of the projected samples within the same classes while maximizing the between-class separability. Meanwhile, an efficient method is developed to solve our formulated problem. Comprehensive analyses, including convergence behavior and parameter determination, together with the relationship to other related approaches, are as well presented. Systematical experiments are conducted to understand the work of our proposed ADA. Promising experimental results on various types of real-world benchmark data sets are provided to examine the effectiveness of our algorithm. Furthermore, we have also evaluated our method in face recognition. They all validate the effectiveness of our method on processing the high-dimensional non-Gaussian data.

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语言: English

文献类型: Article

作者关键词: Adaptive discriminative analysis (ADA); dimensionality reduction; face recognition; high-dimensional non-Gaussian data; linear discriminant analysis (LDA)

KeyWords Plus: FEATURE-EXTRACTION; FACE RECOGNITION; FRAMEWORK; CLASSIFICATION; SUBSPACE

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标题: Semi-Supervised Feature Selection via Insensitive Sparse Regression with Application to Video

Dimension Reduction for Non-Gaussian Data by Adaptive Discriminative Analysis

Tingjin Luo, Chenping Hou, *Member, IEEE*, Feiping Nie, and Dongyun Yi

***Abstract*—High-dimensional non-Gaussian data are ubiquitous in many real applications. Face recognition is a typical example of such scenarios. The sampled face images of each person in the original data space are more closely located to each other than to those of the same individuals due to the changes of various conditions like illumination, pose variation, and facial expression. They are often non-Gaussian and differentiating the importance of each data point has been recognized as an effective approach to process the high-dimensional non-Gaussian data. In this paper, to embed non-Gaussian data well, we propose a novel unified framework named adaptive discriminative analysis (ADA), which combines the sample's importance measurement and subspace learning in a unified framework. Therefore, our ADA can preserve the within-class local structure and learn the discriminative transformation functions simultaneously by minimizing the distances of the projected samples within the same classes while maximizing the between-class separability. Meanwhile, an efficient method is developed to solve our formulated problem. Comprehensive analyses, including convergence behavior and parameter determination, together with the relationship to other related approaches, are as well presented. Systematical experiments are conducted to understand the work of our proposed ADA. Promising experimental results on various types of real-world benchmark data sets are provided to examine the effectiveness of our algorithm. Furthermore, we have also evaluated our method in face recognition. They all validate the effectiveness of our method on processing the high-dimensional non-Gaussian data.**

***Index Terms*—Adaptive discriminative analysis (ADA), dimensionality reduction, face recognition, high-dimensional non-Gaussian data, linear discriminant analysis (LDA).**

I. Introduction

HIGH-DIMENSIONAL non-Gaussian data [1]–[4] are ubiquitous in many real applications, especially in the fields of face recognition [5]–[9]. Due to the effect of the changes of uncontrolled conditions like complex backgrounds, illumination, pose variation, occlusion, and facial expressions, in the original feature space the sampled face images of different persons are located more closely than those of the same individual. If we assume such type of data are Gaussian distributed, the domain of Gaussian variable violates the boundary property. Therefore, face recognition [8], [10], [11] is a typical example of such scenarios. Moreover, face recognition plays an important role in security systems and intelligent surveillance system. It is still a hot topic in the area of machine learning, computer vision and pattern recognition. However, directly processing such high-dimensional non-Gaussian data not only degrades its performance but also is time consuming [12]–[16] in learning tasks. To solve this problem, many dimensionality reduction methods have been introduced for data processing, such as non-negative matrix factorization-based methods [17], [18], k-dimensional coding schemes-based methods [19], and so on. While dimensionality reduction can be applied to both supervised and unsupervised learning, we focus on the problem of supervised learning for non-Gaussian data, where the label information is available.

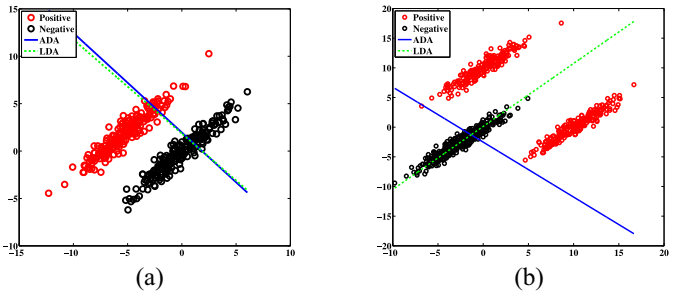

Fig. 1. Dashed line is one projection of LDA, and the solid line is one projection of ADA, for two artificial data sets. (a) Data of each class is sampled from a Gaussian distribution. (b) One class is sampled from a Gaussian distribution, while the other class is sampled from a Gaussian mixture model with two kernels. Note that LDA fails to find the optimal direction in the case with non-Gaussian distributed data.

Although non-Gaussian data arises in many fields, the original data is always assumed as Gaussian distributed in the traditional methods, such as linear discriminative analysis (LDA) [3], [20] and its variants unrelated LDA (ULDA) [21], orthogonal LDA (OLDA) [22], [23], max–min distance analysis [24], and weighted Harmonic mean of trace ratios for multiclass discriminant analysis (WHMDA) [10], [25]. Placing the Gaussian assumption on the non-Gaussian data will discards a great deal of useful structural information [3], [26]. As seen from the toy example

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